$\aleph_0$ Weekly Problem

Ravi Dayabhai Conrad Warren

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Problem

Define a sequence (x_n) as follows:

$$x_0 = 0,\ x_{n+1} = \sqrt{4 + 3x_n}$$

Show that this sequence is convergent and find its limit.

Solution

Claim. *The sequence (x_n) defined by $x_0 = 0$ and $x_{n+1} = \sqrt{4 + 3x_n}$ converges, and its limit is 4.*

Proof. We begin by first establishing that the sequence (x_n) is bounded.

Lemma 1 (Boundedness)**.** *For all $n \geq 0$, then $x_n \leq 4$.*

Proof. We proceed by induction on n. The base case is true because $x_0 = 0 \leq 4$. For the inductive case, we assume $x_n \leq 4$. Then,

$$x_{n+1} = \sqrt{4 + 3x_n} \leq \sqrt{4 + 3 \cdot 4} = \sqrt{16} = 4.$$

By induction, $x_n \leq 4$ for all $n \geq 0$.

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By the Monotone Convergence Theorem, since (x_n) is bounded above (as shown by Lemma 1) and monotonically increasing, it converges to some limit L, which we know to be nonnegative. Notice also that $f(x) = \sqrt{4 + 3x}$ is a continuous function preserving monotonicity (importantly, when x is nonnegative). So, the continuity of f and $L = \lim_{n\to\infty} x_n = \lim_{n\to\infty} x_{n+1}$ gives:

$$L = \lim_{n\to\infty} x_{n+1} = \lim_{n\to\infty} f(x_n) = f\left(\lim_{n\to\infty} x_n\right) = f(L) = \sqrt{4 + 3L}$$

Squaring both sides,

$$L^2 = 4 + 3L \implies L^2 - 3L - 4 = 0 \implies (L - 4)(L + 1) = 0.$$

Therefore, $L = 4$ or $L = -1$. Since $x_n \geq 0$, $L = 4$. □